



**EBN 111/122**  
**ELECTRICITY AND ELECTRONICS**

**Chapter 9**  
**Sinusoidal Steady-State**  
**Analysis**

1



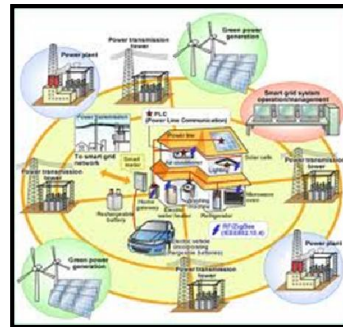
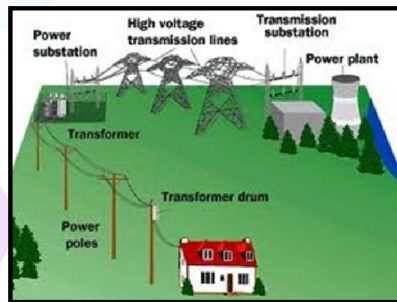
**Sinusoids and Phasor**  
**Chapter 9**

- 9.1 Motivation
- 9.2 Sinusoids' features
- 9.3 Phasors
- 9.4 Phasor relationships for circuit elements
- 9.5 Impedance and admittance
- 9.6 Kirchhoff's laws in the frequency domain
- 9.7 Impedance combinations

2

## 9.1 Motivation

- AC = Alternating Current (Alternates between + and -)
- Vast majority of Power Networks are AC networks because:
  - AC allows for easy stepping up and down of voltages which allows for efficient transfer of energy over long distances.



3

## 9.1 Motivation

### How does one analyse an AC circuit?

#### Method #1: Time domain

KVL:  $-V_s + v_R + v_C + v_L = 0$

$$-10 \cos 4t + iR + \frac{1}{C} \int_{t_0}^t i dt + L \frac{di}{dt} = 0$$

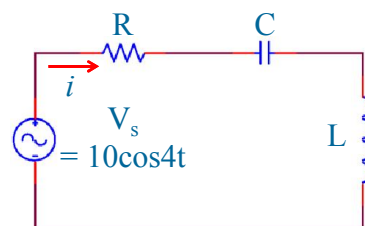
Eliminate  $\int$  by  $\frac{d}{dt} f(t)$ :

$$L \frac{d^2 i}{dt^2} + R \frac{di}{dt} + \frac{i}{C} + 40 \sin 4t = 0$$

2<sup>nd</sup> order differential equation with solutions:

$$i(t) = A_i \cos(4t + \phi_i) \quad V_C(t) = A_C \cos(4t + \phi_C)$$

$$V_R(t) = A_R \cos(4t + \phi_R) \quad V_L(t) = A_L \cos(4t + \phi_L)$$



4

## 9.1 Motivation

### How does one analyse an AC circuit?

#### Method #2:

Frequency or  
**Phasor** domain

KVL:

$$-\bar{V}_s + \bar{V}_R + \bar{V}_C + \bar{V}_L = 0$$

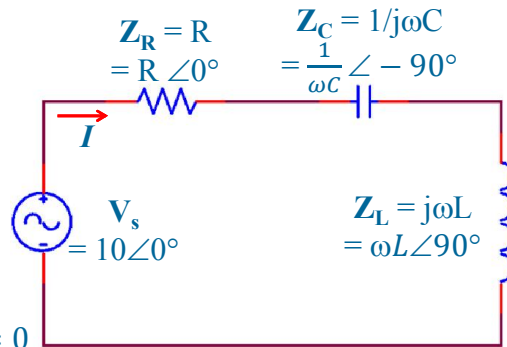
$$-\bar{V}_s + IZ_R + IZ_C + IZ_L = 0$$

$$-10 + I \left( R + \frac{1}{j\omega C} + j\omega L \right) = 0$$

$$I = \frac{10}{R + \frac{1}{j\omega C} + j\omega L} = \frac{10 \angle 0^\circ}{R \angle 0^\circ + \frac{1}{\omega C} \angle -90^\circ + \omega L \angle 90^\circ} = A_i \angle \phi_i^\circ$$

$$A_i \angle \phi^\circ \rightarrow i(t) = A_i \cos(\omega t + \phi_i)$$

5



## 9.2 Sinusoids

- A sinusoid is a signal that has the form of the **sine or cosine** function.
- A general expression for the sinusoid:

$$v(t) = V_m \sin(\omega t + \phi)$$

where

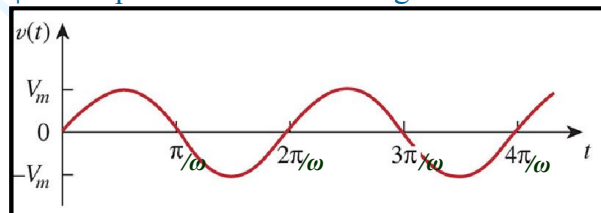
$V_m$  = the **amplitude** of the sinusoid

$\omega$  = the angular frequency in radians/s

$t$  = time in seconds

$\omega t$  = argument of the sinusoid in radians

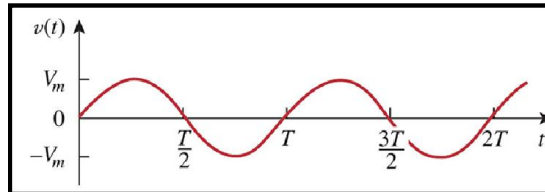
$\phi$  = the phase in radians or degrees



6

## 9.2 Sinusoids

### V vs Time (not $\omega t$ )



$T$  = period [s]

At  $t = T$ :  $\omega T = 2\pi$  :

$$T = \frac{2\pi}{\omega}$$

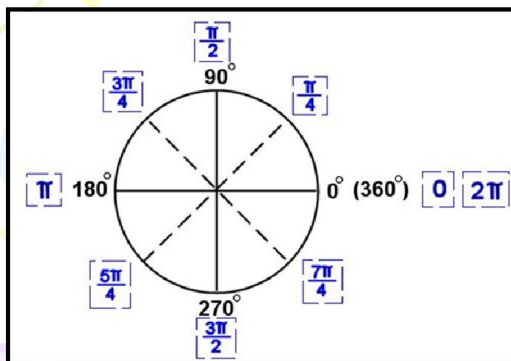
$f = \frac{1}{T}$  frequency in [Hz]

$\omega = 2\pi f$  [rad/s]

7

## 9.2 Sinusoids

### Radians $\leftrightarrow$ Degrees



$$360^\circ = 2\pi \text{ rad}$$

$$180^\circ = \pi \text{ rad}$$

Conversion factors:

$$\text{deg} \rightarrow \text{rad}: \theta^\circ \times \frac{\pi}{180^\circ} = \phi \text{ rad}$$

$$\text{rad} \rightarrow \text{deg}: \phi \text{ rad} \times \frac{180^\circ}{\pi} = \theta^\circ$$

8

